

Module Summary & Interpretation

Wine Quality Data Workflow

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AI Programming Fundamentals Capstone Module

Overview

This project implements a complete, reproducible data workflow on the UCI Wine Quality dataset using Python, Pandas, NumPy, Matplotlib, and Seaborn inside a single Jupyter Notebook (*data_workflow.ipynb*). The workflow ingests the red and white wine data, cleans and de-duplicates it, explores it with reusable functions, and visualizes how chemical properties relate to expert quality scores. The dataset used is the Wine Quality dataset from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/186/wine+quality>). Structuring the analysis as an executable, well-documented notebook follows established guidance for reproducible data science (Danchev, 2022).

Dataset Description

The Wine Quality dataset describes Portuguese “Vinho Verde” wines and was originally compiled to model wine preferences from physicochemical tests (Cortez et al., 2009). It is distributed as two files: 1,599 red wine samples and 4,898 white wine samples, which I combined into a single table of 6,497 rows and added a *wine_type* column to preserve each sample’s origin. After removing exact duplicate rows, 5,320 unique samples remained. Each record contains eleven numeric physicochemical features (for example fixed acidity, volatile acidity, residual sugar, chlorides, density, pH, sulphates, and alcohol) and an integer *quality* score from 0 to 10 that represents the median of at least three evaluations made by wine experts. My analysis focused on the *quality* score as the target of interest and on *alcohol*, *density*, and *volatile acidity* as the features most strongly associated with it.

Workflow Description

Ingestion. The two semicolon-separated CSV files were read with Pandas, each tagged with its wine type, and concatenated into one DataFrame so both varieties could be analyzed together.

Cleaning. I wrote small, documented functions to standardize column names (lowercasing and replacing spaces with underscores), drop exact duplicate rows, and report missing values using Pandas’ missing-data tools (The pandas development team, 2024). The completeness check confirmed there were no missing values to impute.

Exploratory analysis. Reusable EDA functions produced summary statistics, grouped views (quality by wine type and alcohol by quality score), and a correlation ranking of every feature against quality.

Visualizations. Four labeled Matplotlib/Seaborn charts were created: a distribution of quality scores by wine type, a boxplot of alcohol across quality scores, a correlation heatmap, and a bar chart of average quality by wine type.

Summary. A final Markdown section in the notebook records what was learned, the key patterns, and the limitations of the analysis, and this report expands on that interpretation with supporting citations.

Key Decisions and Assumptions

Cleaning choices. I removed 1,177 exact duplicate rows because identical physicochemical readings with identical scores most likely represent repeated records that would otherwise inflate apparent sample size and bias summary statistics; documenting and reporting this step keeps the decision transparent and reversible, consistent with good-enough practices for data management (Wilson et al., 2017). Standardizing column names reduces the risk of subtle bugs from inconsistent naming. I assumed that exact duplicates were redundant rather than genuinely distinct wines that happened to share identical measurements.

EDA focus. Because the analytical question is how chemistry relates to perceived quality, the EDA centered on the correlation between each feature and the quality score and on grouped comparisons across wine type and quality level. This mirrors the original study, which modeled quality directly from physicochemical inputs (Cortez et al., 2009).

Plot design. Figure 1 (count of wines by quality and type) was designed to show the distribution and class imbalance of the target. Figure 2 (boxplot of alcohol by quality) was designed to reveal the trend and spread of the strongest correlate. Figure 3 (correlation heatmap) was designed to show the full feature-to-feature and feature-to-target structure at a glance. Figure 4 (average quality by wine type) was designed to compare the two varieties directly. Each chart carries a descriptive title and labeled axes so it can be read on its own, supporting the transparent communication of results emphasized in reproducible data science workflows (Danchev, 2022).

Results and Interpretation

The quality scores are concentrated at 5 and 6 and are roughly bell-shaped, with very few wines at the extremes (Figure 1). Alcohol content is the single strongest positive correlate of quality (Pearson $r \approx +0.47$), and the boxplot shows median alcohol rising clearly for wines rated 6 and above (Figure 2). Conversely, density ($r \approx -0.33$) and volatile acidity ($r \approx -0.27$) are the most negatively associated features, indicating that lighter, less-acetic wines tend to score higher (Figure 3). At the group level, white wines have a marginally higher average quality score (≈ 5.86) than red wines (≈ 5.62) (Figure 4). These associations are consistent with prior modeling of this dataset, in which physicochemical inputs including alcohol carried meaningful signal about sensory preference (Cortez et al., 2009).

Responsible Practice (Bias and Data Quality)

Several handling choices could introduce bias. Dropping duplicate rows assumes those records are redundant; if any were in fact distinct wines, removal would subtly distort the distribution, so I reported the exact count removed to make the effect auditable. The dataset is also strongly imbalanced toward mid-range scores, which means any pattern at the high or low end rests on relatively few samples and should be treated cautiously. Treating the ordinal quality score as a numeric value is a simplification that can overstate the precision of correlations. Finally, the data covers only Portuguese Vinho Verde wines, so conclusions may not generalize to other regions or styles (Cortez et al., 2009). To reduce these risks I documented every transformation in code, kept the raw files unmodified, and would next analyze red and white wines separately and use rank-based or ordinal methods to confirm the relationships; transparent, documented data management is itself a recognized safeguard against irreproducible or misleading results (Wilson et al., 2017).

Reproducibility

Anyone can rerun this work. The project pins its dependencies in *requirements.txt*, which is installed with *pip install -r requirements.txt*, and the analysis lives entirely in *data_workflow.ipynb*, which runs top to bottom against the CSV files included in the *data/* folder. The project is version-controlled with Git: incremental commits capture each stage of the workflow (loading, cleaning, EDA, visualization, and summary), and a separate feature branch was used to develop and review changes before merging into the main branch. Combining an executable notebook, an explicit environment specification, and disciplined version control is the core recipe for reproducible data science advocated by Danchev (2022) and reinforced by broader scientific-computing guidance (Wilson et al., 2017).

References

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